Trapping Light in Diamond: The NV Center as a Room-Temperature Qubit

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DFT confirms NV⁻ as the thermodynamically stable, spin-triplet charge state in diamond, with a computed ZPL consistent with optical spin-readout for quantum sensing.

Why Diamond?

- Superconducting qubits need near-zero cooling — costly, hard to scale
- Point defects in wide-bandgap crystals offer a room-temperature alternative
- NV center forms when, inside diamond's carbon lattice, nitrogen replaces one carbon atom and a carbon is removed, leaving an vacancy.

NV center is a spin qubit that can as quibt and nanoscale sensor, this project focuses on the properties enabling the optical spin-readout.

Requirement	Why it matters	Method
Optically active	Can interact with light — enables initialization and readout	ZPL calculation
Highly localized states	Individually addressable — a single, well-defined qubit	LDOS
Levels inside the band gap	Isolated from bulk bands — accessible, distinguishable transitions	Formation energy
Paramagnetic ground state	Nonzero spin — enables spin-based control and readout	Kohn-Sham diagram
Thermodynamically stable	Ensures the defect actually forms and persists in that charge state	Formation energy

Building the Defect, Step by Step

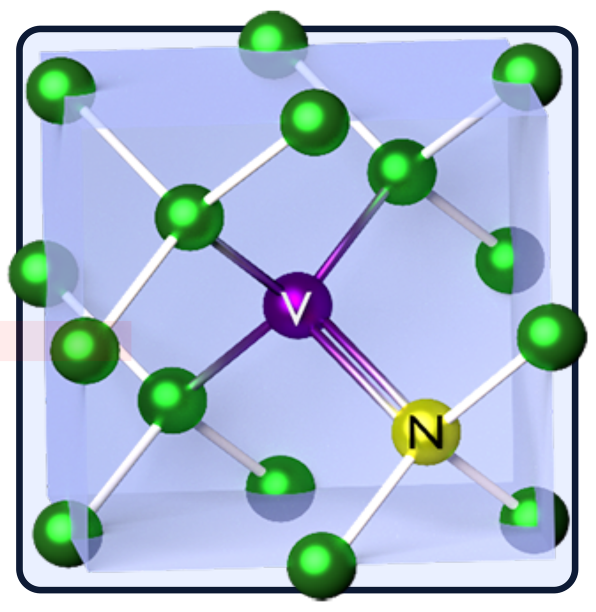


Figure 1: NV center in diamond

1 Bulk Validation
Evaluate lattice constant
PBE vs. HSE06

2 Defect Construction
Build NV supercell
(N substitution + C vacancy)

3 Density of States
Identify in-gap defect states
Localization

4 Kohn-Sham States
Determine level occupations

5 Thermodynamics
Calculate formation energies
Transitions levels

6 Optical Properties
Compute zero phonon line
Wavelength

Which Charge Is More Stable?

To find the thermodynamically favorable charge state, we compute NV's formation energy across all charges (q = −3 to +2) vs. Fermi level:

$$E = E^q_{def} - E_{perf} + 2\mu_C - \mu_N + q(E_{VBM} + E_F) + E_{corr}$$

total energies defect vs perfect cell | atomic chemical potentials | electron reservoir term | finite-size charge correction

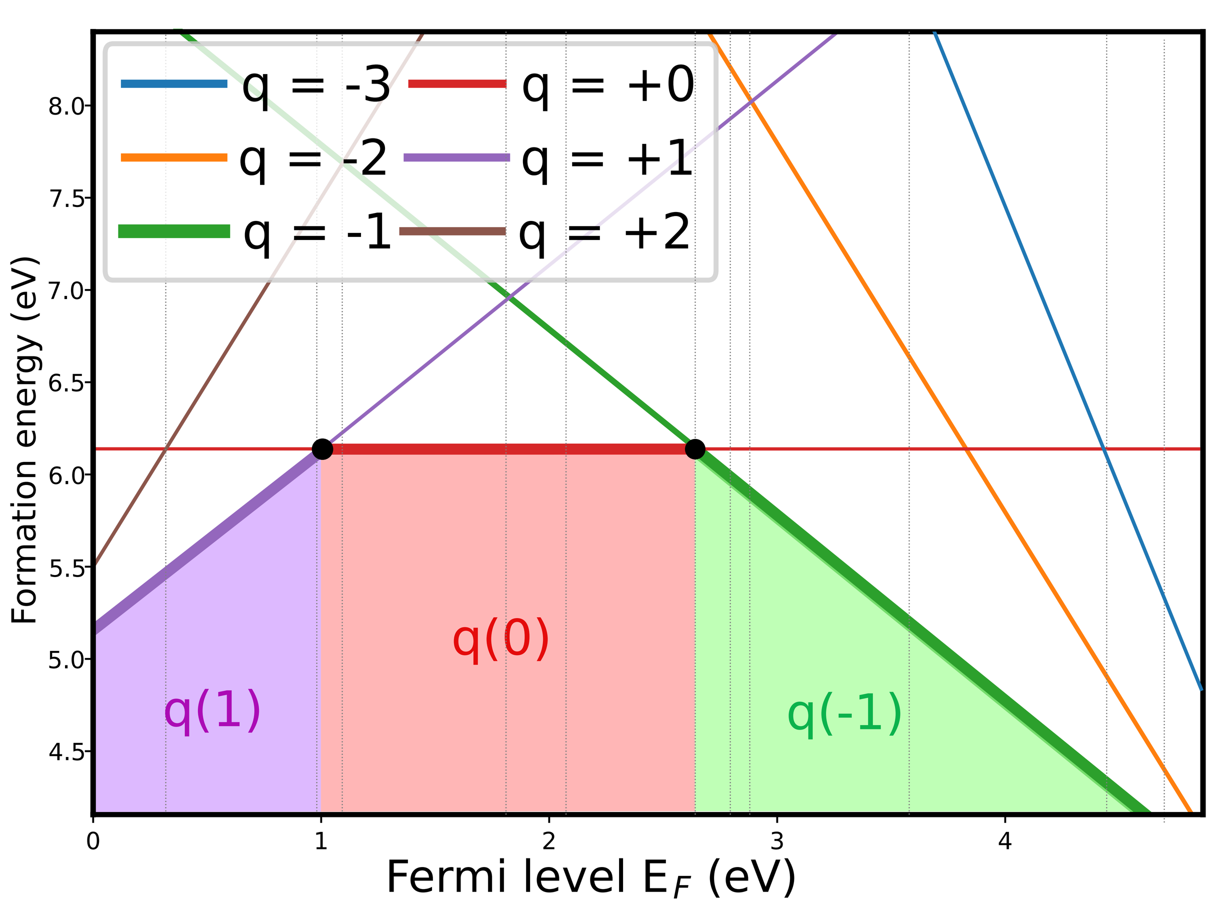


Figure 2: NV⁻ is the lowest-formation-energy charge state across most of the gap (E_F > 2.64 eV) — including the range typical of N-doped diamond — confirming it as the thermodynamically stable.

How NV Reacts to Light

Three charge states dominate, q = +1, 0, −1, with spin and occupation determining which is qubit-relevant.

Charge	Spin (S)	Unpaired e⁻
+1	0.0	0
0	0.5	1
−1	**1.0**	**2**

Table 1: NV⁻ is the only relevant charge state with a spin-triplet S=1 ground state, two unpaired electrons make optical spin-readout possible.

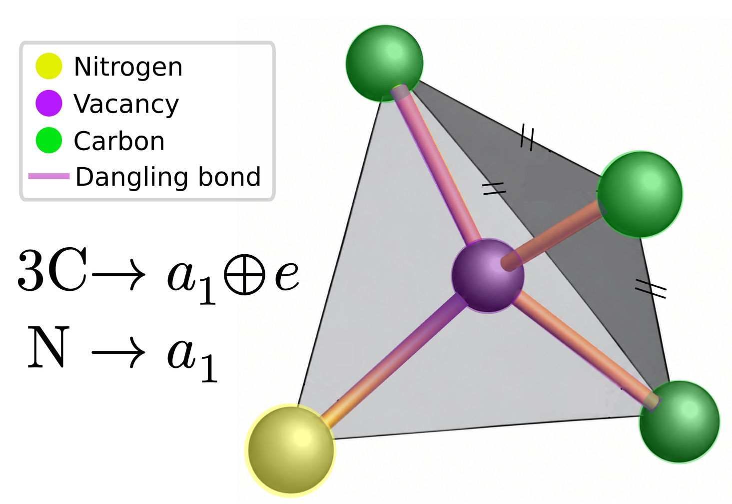
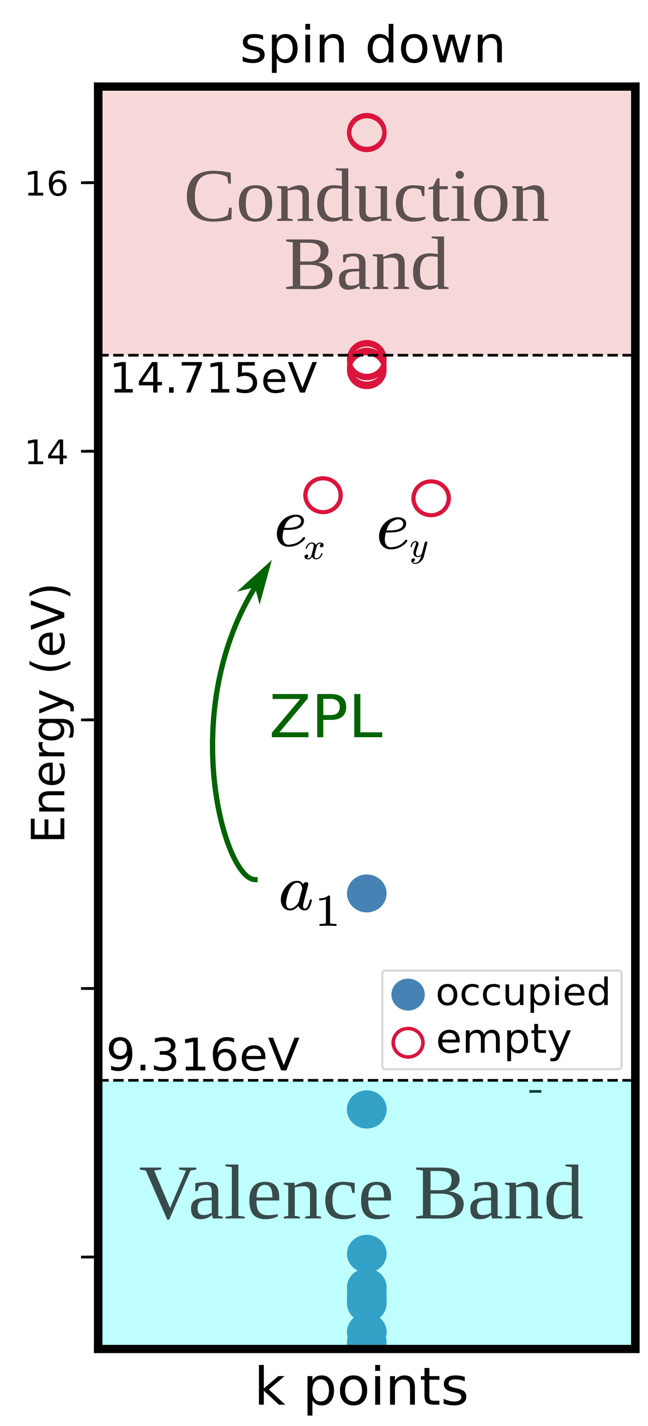


Figure 3: 120° rotation about the N–vacancy axis (C_{3v}) maps the three C dangling bonds onto each other, splitting them into a_1 and e.

Figure 4: One electron moves from a_1 (filled) to the doubly-degenerate e level, driving the $^3A_2 \rightarrow {}^3E$ optical transition



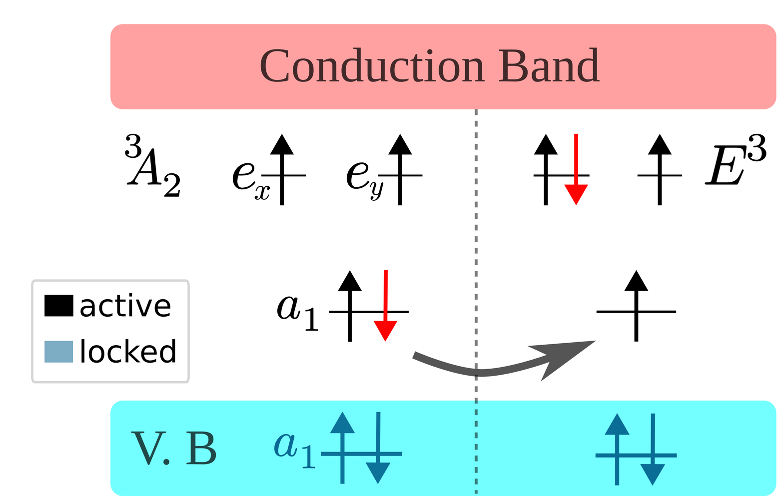


Figure 5: Of the two a_1 combinations from C_{3v} symmetry, one merges into the valence band (locked, always filled); the other remains in-gap, active alongside the e pair.

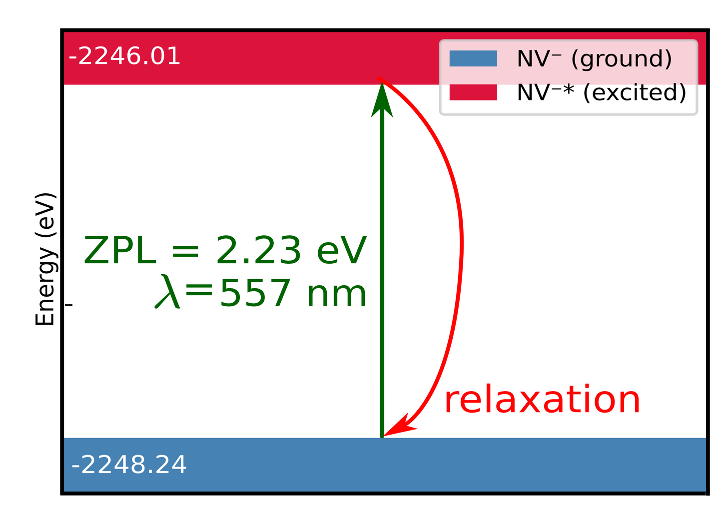


Figure 6: Computed ZPL lies in the visible spectrum, confirming NV⁻'s optical transition can be addressed with visible light, basis for optical spin-readout.

Can We Trust the Method?

Quantity	PBE	HSE06	Exp
Lattice constant (Å)	3.5737	3.5481	3.567
Band gap (eV)	4.12	5.34	5.47
ZPL (eV)		**2.23**	**1.945**

Table 2: HSE06 corrects PBE's band-gap underestimation, justifying its use for defect calculations

Quantity (eV)	This work	Literature
transition level $\varepsilon(0+)$	0.981	0.9–1.2
transition level $\varepsilon(0/-)$	2.64	≈2.7–2.8

Table 3: Formation energies depend on the chemical potential reference used and are best compared within the same convention.

Confirm the DFT setup reproduces known diamond physics and that the NV-specific quantities are internally consistent.

What This Means For Quantum Sensing

NV⁻ is both thermodynamically stable and optically addressable — properties already exploited in real-world NV-diamond quantum sensors — and this work provides first-principles confirmation of why NV⁻ is the charge state responsible.

- Point defects like the NV center are emerging as a strong platform bridging quantum sensing, computation, and scalable room-temperature devices, future works includes long coherence time calcs.

References

- Dreyer et al., Annu. Rev. Mater. Res. 48, 1 (2018).
- Panana Vera, J. I., Thesis, UNMSM
- Doherty et al., Phys. Rep. 528, 1 (2013).

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